EXPERIMENT 10.

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1:

Set Up Your HTML File First, create an HTML file to hold your canvas for the chart and include Chart.js.

HTML

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Inventory Management Visualization</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <h1>Inventory Management System</h1>
  <p class="subtitle">Stock distribution across product categories</p>
  <div class="stats-grid">
    <div class="stat-card">
      <div class="stat-num">580</div>
      <div class="stat-lbl">Total Items</div>
    </div>
    <div class="stat-card">
      <div class="stat-num">5</div>
      <div class="stat-lbl">Categories</div>
    </div>
    <div class="stat-card">
      <div class="stat-num">200</div>
      <div class="stat-lbl">Max Stock</div>
    </div>
    <div class="stat-card">
      <div class="stat-num">50</div>
      <div class="stat-lbl">Min Stock</div>
    </div>
  </div>
  <div class="charts-grid">
```

```html
    <div class="chart-card">
      <h2>Inventory Distribution</h2>
      <div class="legend" id="pieLegend"></div>
      <div class="canvas-wrap">
        <canvas id="pieChart"
          role="img"
          aria-label="Pie chart: Electronics 200, Clothing 150, Home
Appliances 100, Books 80, Toys 50">
          Inventory by category: Electronics 200, Clothing 150, Home
Appliances 100, Books 80, Toys 50.
        </canvas>
      </div>
    </div>
    <div class="chart-card">
      <h2>Items in Stock by Category</h2>
      <div class="canvas-wrap tall">
        <canvas id="barChart"
          role="img"
          aria-label="Bar chart: Electronics 200, Clothing 150, Home
Appliances 100, Books 80, Toys 50">
          Stock counts: Electronics 200, Clothing 150, Home Appliances 100,
Books 80, Toys 50.
        </canvas>
      </div>
    </div>

  </div>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <script src="script.js"></script>
</body>
</html>
```

CSS

```css
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}
body {
  font-family: Arial, sans-serif;
  background-color: #f4f6fb;
  text-align: center;
  padding: 40px 20px;
  color: #1a1a2e;
```

```
}
h1 {
  font-size: 28px;
  font-weight: 700;
  color: #1a1a2e;
  margin-bottom: 6px;
}
.subtitle {
  font-size: 14px;
  color: #666;
  margin-bottom: 30px;
}
.stats-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(120px, 1fr));
  gap: 14px;
  max-width: 700px;
  margin: 0 auto 36px;
}
.stat-card {
  background: #ffffff;
  border-radius: 10px;
  padding: 16px 10px;
  border: 1px solid #ede9ff;
}
.stat-num {
  font-size: 26px;
  font-weight: 800;
  color: #7c3aed;
}
.stat-lbl {
  font-size: 12px;
  color: #888;
  margin-top: 4px;
}
.charts-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
  gap: 24px;
  max-width: 900px;
  margin: 0 auto;
}
.chart-card {
  background: #ffffff;
  border-radius: 14px;
  padding: 20px;
  border: 1px solid #e8e8e8;
  text-align: left;
```

```
}
.chart-card h2 {
  font-size: 15px;
  font-weight: 700;
  color: #333;
  margin-bottom: 14px;
  text-align: center;
}
.legend {
  display: flex;
  flex-wrap: wrap;
  justify-content: center;
  gap: 10px;
  margin-bottom: 14px;
}
.legend-item {
  display: flex;
  align-items: center;
  gap: 5px;
  font-size: 11px;
  color: #555;
}
.legend-dot {
  width: 10px;
  height: 10px;
  border-radius: 2px;
  flex-shrink: 0;
}
.canvas-wrap {
  position: relative;
  height: 280px;
  width: 100%;
}
.canvas-wrap.tall {
  height: 320px;
}
canvas {
  display: block;
}
```

JAVASCRIPT

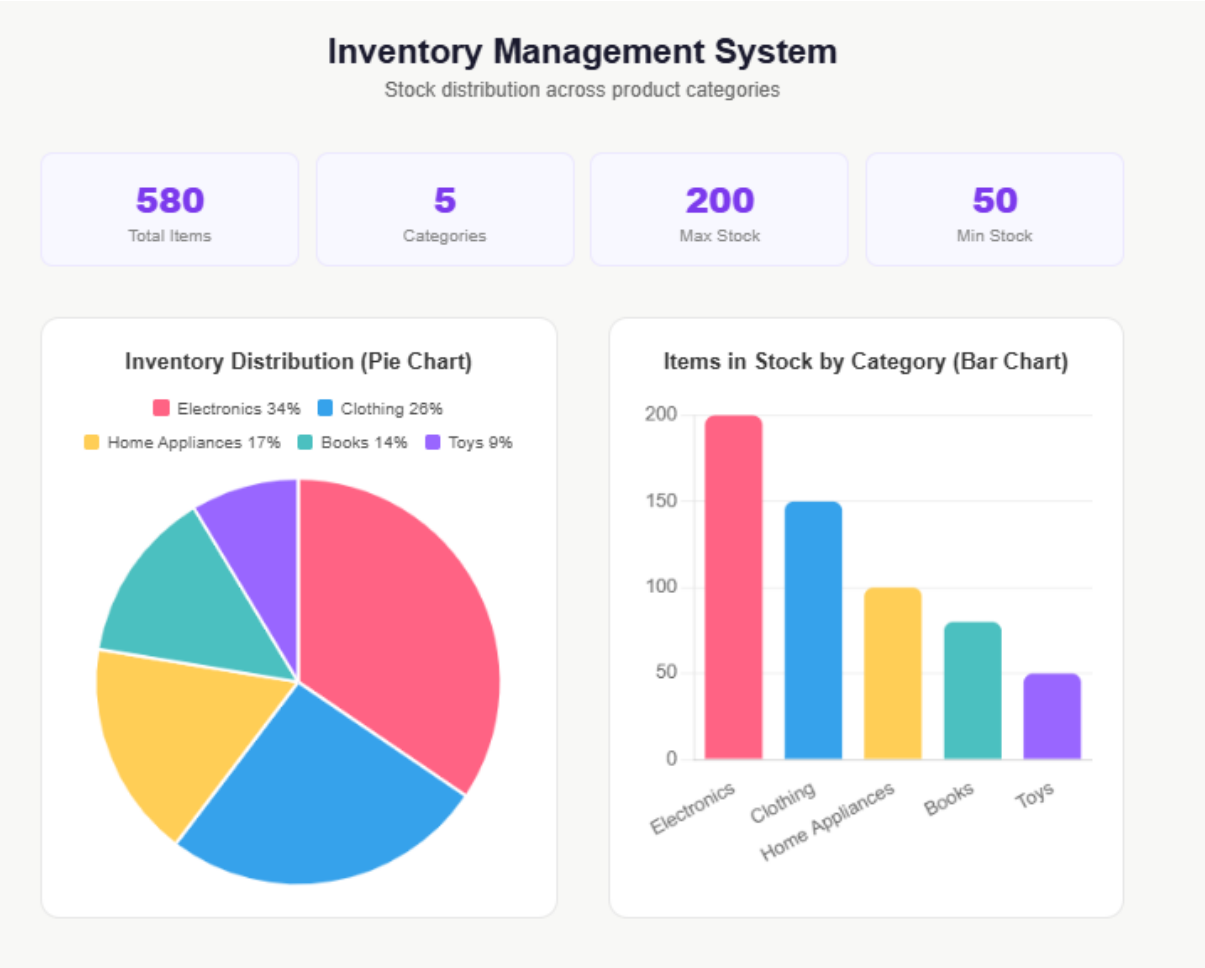
```
var labels = ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'];
var stockData = [200, 150, 100, 80, 50];
var colors = ['#FF6384', '#36A2EB', '#FFCE56', '#4BC0C0', '#9966FF'];
var total = stockData.reduce(function (sum, val) { return sum + val; }, 0);
```

```
var legendContainer = document.getElementById('pieLegend');
labels.forEach(function (label, i) {
  var pct = Math.round((stockData[i] / total) * 100);
  var item = document.createElement('span');
  item.className = 'legend-item';
  item.innerHTML =
    '<span class="legend-dot" style="background:' + colors[i] + '"></span>' +
    label + ' ' + pct + '%';
  legendContainer.appendChild(item);
});
var ctxPie = document.getElementById('pieChart').getContext('2d');
var pieChart = new Chart(ctxPie, {
  type: 'pie',
  data: {
    labels: labels,
    datasets: [{
      label: 'Items in Stock',
      data: stockData,
      backgroundColor: colors,
      borderColor: '#ffffff',
      borderWidth: 2
    }]
  },
  options: {
    responsive: true,
    maintainAspectRatio: false,
    plugins: {
      legend: {
        display: false
      },
      title: {
        display: false
      },
      tooltip: {
        callbacks: {
          label: function (ctx) {
            var pct = Math.round(ctx.parsed / total * 100);
            return ' ' + ctx.label + ': ' + ctx.parsed + ' items (' + pct +
'%)';
          }
        }
      }
    }
  }
});
var ctxBar = document.getElementById('barChart').getContext('2d');
var barChart = new Chart(ctxBar, {
  type: 'bar',
```

```
data: {
  labels: labels,
  datasets: [{
    label: 'Items in Stock',
    data: stockData,
    backgroundColor: colors,
    borderRadius: 6,
    borderWidth: 0
  }]
},
options: {
  responsive: true,
  maintainAspectRatio: false,
  plugins: {
    legend: {
      display: false
    },
    title: {
      display: false
    },
    tooltip: {
      callbacks: {
        label: function (ctx) {
          return ' Items in stock: ' + ctx.parsed.y;
        }
      }
    }
  },
  scales: {
    y: {
      beginAtZero: true,
      ticks: {
        stepSize: 50
      },
      grid: {
        color: 'rgba(0, 0, 0, 0.06)'
      }
    },
    x: {
      grid: {
        display: false
      },
      ticks: {
        autoSkip: false,
        maxRotation: 20
      }
    }
  }
```

```
  }
});
```

OUTPUT


Inventory Management System
Stock distribution across product categories
580
Total Items
5
Categories
200
Max Stock
50
Min Stock
Inventory Distribution (Pie Chart)
Electronics 34%
Clothing 26%
Home Appliances 17%
Books 14%
Toys 9%
Items in Stock by Category (Bar Chart)
200
150
100
50
0
Electronics
Clothing
Home Appliances
Books
Toys